

# JANICE BENITA F

## AI/ML ENGINEER | SOFTWARE ENGINEER

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### PROFESSIONAL SUMMARY

Information Technology undergraduate with internship experience in machine learning, computer vision, data analytics, and software engineering. Built explainable deep-learning systems, AI agents, REST APIs, ML workflows, dashboards, and Docker-based verification using Python, PyTorch, OpenCV, FastAPI, SQL, React, TypeScript, and CI/CD.

### EDUCATION

**St. Joseph's College of Engineering** 2023–2027  
*Bachelor of Technology in Information Technology | CGPA: 8.30/10.00*  
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning

### TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Computer Vision, Explainable AI, Feature Engineering, Model Evaluation, PyTorch, scikit-learn, OpenCV  
Software: Python, FastAPI, Flask, REST APIs, SQL, SQLite, MySQL, Pytest, React, TypeScript, JavaScript  
Data & Delivery: Pandas, NumPy, Plotly, Power BI, Tableau, Streamlit, Git, GitHub Actions, Docker, Ruff, MyPy, Bandit

### EXPERIENCE

**Larsen & Toubro, Chennai** May 2025 – Jul 2025  
*Data Analytics and Machine Learning Intern*

- Developed a ResNet-18 and Grad-CAM concrete-crack classification pipeline, achieving 95.7% accuracy.
- Analyzed 10,000+ inspection records using Python and SQL, reducing inspection-analysis turnaround from approximately 25–30 days to 3–5 days; developed infrastructure-quality dashboards.

**YuvaIntern** Mar 2026 – May 2026  
*Machine Learning and Data Analytics Intern*

- Performed exploratory analysis, feature engineering, predictive modeling, model evaluation, and visualization on agricultural data; built reusable workflows using Python and Pandas.

**CodeBind Technologies** Jul 2024 – Aug 2024  
*Web Development Intern*

- Developed responsive applications using HTML, CSS, and JavaScript; integrated REST APIs and improved frontend usability.

### SELECTED PROJECTS

**SentinelOps – Autonomous AI Reliability Engineer**  
*Python, FastAPI, React, TypeScript, SQLite, Docker*

- Built an evidence-first agent that ranks root-cause hypotheses, reproduces failures, proposes bounded patches, and runs deterministic Local/Docker verification. Human approval is required before preparing a pull-request record; the system does not deploy automatically.

**EndoXAI – Explainable Root Canal Treatment Prediction**  
*Deep Learning, Grad-CAM, Hugging Face Spaces*

- Developed an explainable clinical decision-support research prototype as AI/ML Software Developer and Co-Author; poster won Second Prize at IDDC 2026, postgraduate category.

**Digital Quality Intelligence Platform**  
*Python, Analytics, Dashboards*

- Developed modular Six Sigma, Cp/Cpk, anomaly-detection, supplier-benchmarking, ACI-compliance, and executive-dashboard workflows; associated paper accepted in IJATCSE 15(4), 2026.

**Explainable Concrete Crack Detection**  
*ResNet-18, Grad-CAM, OpenCV*

- Built an explainable classifier with 95.7% experimental accuracy; associated research published in IJERT in 2026.

### RESEARCH AND RECOGNITION

Accepted: Design and Implementation of the Digital Quality Intelligence Platform: A Modular Decision-Support Software Framework for Engineering Quality Intelligence. IJATCSE, 15(4), 2026.  
Published: Explainable Deep Learning Framework for Automated Concrete Crack Detection Using ResNet-18. IJERT, 2026. DOI: 10.5281/zenodo.20124334  
Recognition: Second Prize, International Digital Dentistry Congress 2026 Poster Presentation, Postgraduate Category; Runner-up, CodeVerse Inter-Collegiate Technical Symposium.